

APUNTES DE CLASE DE MECANICA CUANTICA

TEMA: PROBLEMA DE DOS CUERPOS

1. EL PROBLEMA DE LOS DOS CUERPOS

Consideremos el hamiltoniano de dos partículas que interactúan por medio de un potencial

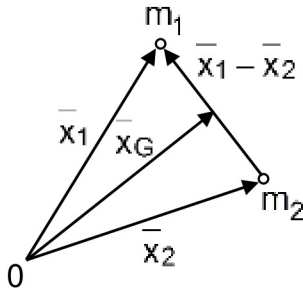
$V(\bar{t}, \bar{x}_1, \bar{x}_2)$:

$$H = \frac{1}{2m_1} P_1^2 + \frac{1}{2m_2} P_2^2 + V \quad (1)$$

$$H\psi + \frac{\hbar}{i} \partial\psi = 0 \quad (2)$$

$$(V\psi)(\bar{t}, \bar{x}_1, \bar{x}_2) = V(\bar{t}, \bar{x}_1, \bar{x}_2) \psi(\bar{t}, \bar{x}_1, \bar{x}_2) \quad (3)$$

$$\text{Consideremos: } V(\bar{t}, \bar{x}_1, \bar{x}_2) = \underbrace{\tilde{V}(\bar{t}, \bar{x}_1 - \bar{x}_2)} \quad (4)$$



$$\bar{x} = \bar{x}_1 - \bar{x}_2$$

$$\bar{x}_G = \frac{m_1 \bar{x}_1 + m_2 \bar{x}_2}{m_1 + m_2} \quad (5)$$

$$\phi(t, \bar{x}, \bar{x}_G) = \psi(t, \bar{x}_1, \bar{x}_2) \quad (6)$$

Construiremos los correspondientes operadores, de posición y de momentum.

$$\bar{x} = (x_1, x_2, x_3): \text{variable relativa} \quad (7)$$

$$\bar{x}_k = (x_{k1}, x_{k2}, x_{k3}): \text{variable de la partícula k-ésima} \quad (8)$$

$$\bar{x}_G = (x_{G1}, x_{G2}, x_{G3}): \text{variable el CM} \quad (9)$$

Usaremos las siguientes notaciones:

$$D_j F = \frac{\partial F}{\partial x_j} \quad (10)$$

$$\bar{D}^2 F = D_1^2 F + D_2^2 F + D_3^2 F \quad (11)$$

$$D_{G_j} F = \frac{\partial F}{\partial x_{G_j}} \quad (12)$$

$$D_{kj} F = \frac{\partial F}{\partial x_{kj}} \quad (13)$$

$$D_j^2 F = \frac{\partial^2 F}{\partial x_j^2} \quad (14)$$

$$\partial F = \frac{\partial F}{\partial t} \quad (15)$$

$$\bar{D} \equiv (D_1, D_2, D_3) \quad (16)$$

$$\bar{D}_1 \equiv (D_{11}, D_{12}, D_{13}) \quad (17)$$

- Verificar:

$$\bar{x}_1 = \bar{x}_G + \frac{m_2}{m} \bar{x} \quad (18)$$

$$\bar{x}_2 = \bar{x}_G - \frac{m_1}{m} \bar{x} \quad (19)$$

- $D_{1k} x_j = \delta_{kj} ; D_{2k} x_j = -\delta_{kj}$ (20)

$$x_j = x_{1j} - x_{2j} \quad (21)$$

- $D_{1k} x_j = D_{1k} x_{1j} - \cancel{D_{1k} x_{2j}} = \delta_{kj}$ (22)

- $D_{2k} x_j = \cancel{D_{2k} x_{1j}} - D_{2k} x_{2j} = -\delta_{kj}$ (23)

- $D_{1k} x_{Gj} = \frac{m_1}{m} \delta_{kj}$ (24)

- $D_{2k} x_{Gj} = \frac{m_2}{m} \delta_{kj}$

$$\rightarrow x_{Gj} = \frac{m_1 x_{1j} + m_2 x_{2j}}{m_1 + m_2} \quad (25)$$

- $D_{1k} x_{Gj} = \frac{m_1}{m} D_{1k} x_{1j} = \frac{m_1}{m} \delta_{kj}$ (26)

- Verificar:

$$D_{1k} \psi = D_k \phi + \frac{m_1}{m} D_{Gk} \phi \quad (27)$$

$$\phi(t, \bar{x}, \bar{x}_G) = \psi(t, \bar{x}_1, \bar{x}_2)$$

$$D_{1k} \psi = \underbrace{D_j \phi D_{1k} x_j}_{\text{}} + D_{Gj} \phi \underbrace{D_{1k} x_{Gj}}_{\text{}}$$

$$D_{1k}\psi$$

$$= D_k\phi + \frac{m_1}{m} D_{Gk}\phi$$

$$\bullet \quad D_2 \psi = -D_k\phi + \frac{m_2}{m} D_{Gk}\phi \quad (28)$$

$$\bullet \quad \bar{D}_1^2 \psi = \bar{D}^2 \phi + \left(\frac{m_1}{m}\right)^2 \bar{D}_G^2 \phi + 2 \frac{m_1}{m} \sum_j D_j D_{Gj} \phi \quad (29)$$

$$D_{1k}\psi = D_k\phi + \frac{m_1}{m} D_{Gk}\phi$$

$$D_{1k}^2 \psi = D_j D_k \phi D_{1k} x_j + D_{Gj} D_k \phi D_{1k} x_{Gj}$$

$$+ \frac{m_1}{m} D_j D_{Gk} \phi D_{1k} x_j + \frac{m_1}{m} D_{Gj} D_{Gk} \phi D_{1k} x_{Gj}$$

$$= D_k^2 \phi + \frac{m_1}{m} D_{Gk} D_k \phi + \frac{m_1}{m} D_k D_{Gk} \phi + \left(\frac{m_1}{m}\right)^2 D_{Gk}^2 \phi$$

$$\rightarrow \bar{D}_1^2 \psi = \bar{D}^2 \phi + \left(\frac{m_1}{m}\right)^2 \bar{D}_G^2 \phi + \frac{2m_1}{m} \sum_k D_{Gk} D_k \phi$$

Tarea:

$$\bar{D}_{2\psi}^2 = \bar{D}^2 \phi + \left(\frac{m_2}{m} \right)^2 \bar{D}_G^2 \phi - \frac{2m_2}{m} \sum_j D_{Gj} D_j \phi \quad (30)$$

$$\frac{\bar{D}_1^2 \psi}{m_1} + \frac{\bar{D}_2^2 \psi}{m_2} = \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \bar{D}^2 \phi + \frac{\bar{D}_G^2 \phi}{m} \quad (31)$$

$$\frac{1}{\mu} \equiv \frac{1}{m_1} + \frac{1}{m_2} \quad (\text{masa reducida}) \quad (32)$$

$$\frac{\bar{D}_1^2}{m_1} \psi + \frac{\bar{D}_2^2}{m_2} \psi = \frac{\bar{D}^2 \phi}{\mu} + \frac{\bar{D}_G^2 \phi}{m} \quad (33)$$

$$\therefore \boxed{\frac{\bar{P}_1^2}{2m_1} + \frac{\bar{P}_2^2}{2m_2} = \frac{\bar{P}^2}{2\mu} + \frac{\bar{P}_G^2}{2m}} \quad (34)$$

$$H\phi + \frac{\hbar}{i} \partial \phi = 0 \quad (35)$$

$$H = \frac{\bar{P}^2}{2\mu} + \frac{1}{2m} \bar{P}_G^2 + V \quad (36)$$

$$\phi(t, \bar{x}, \bar{x}_G) = X(\bar{x}) Y(t, \bar{x}_G) \quad (37)$$

Variables separables

$$HXY + \frac{\hbar}{i} \partial XY = 0 \quad (38)$$

$$\frac{1}{XY} \left(\frac{\bar{P}^2}{2\mu} XY + \frac{\bar{P}_G^2}{2m} XY + VXY + \frac{\hbar}{i} \partial XY \right) = 0 \quad (39)$$

$$\underbrace{\left(\frac{1}{X} \frac{P^2 X}{2\mu} + \frac{VX}{X} \right)}_E + \underbrace{\left(\frac{1}{Y} \frac{P_G^2 V}{2m} + \frac{\hbar}{i} \frac{\partial Y}{Y} \right)}_{-E} = 0 \quad (40)$$

$$\boxed{\frac{P^2 X}{2\mu} + V X = E X} \quad (41)$$

Corresponde al movimiento relativo de los dos partículas.

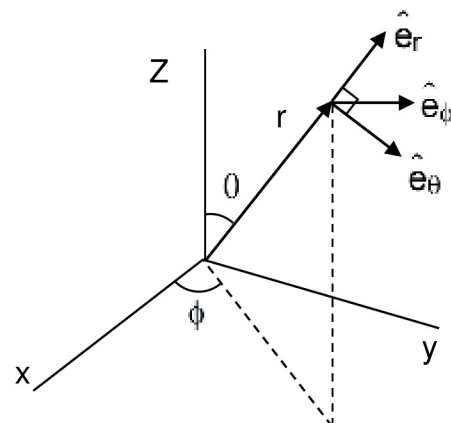
$$\frac{P_G^2 Y}{2m} + E Y + \frac{\hbar}{i} \frac{\partial Y}{Y} = 0 \quad (42)$$

$$\rightarrow Y = n = e^{i(\vec{k} \cdot \vec{x}_G - \omega t)} \quad (43)$$

$$\wedge \hbar \omega = \frac{k^2 \hbar^2}{2m} + E \quad (44)$$

2. SOLUCION DEL PROBLEMA DE DOS CUERPOS EN COORDENADAS ESFERICAS

Dado que el potencial solo depende del módulo de la variable posición, es conveniente considerar los operadores en coordenadas esféricas



$$\bar{x} = \bar{r} = r(\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta) \quad (2.1)$$

$$\mathbf{e}_r = \frac{\partial \bar{r} / \partial r}{\| \quad \|} \quad \mathbf{e}_\theta = \frac{\partial \bar{r} / \partial \theta}{\| \quad \|} \quad \mathbf{e}_\phi = \frac{\partial \bar{r} / \partial \phi}{\| \quad \|} \quad (2.2)$$

$$\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{e}_\phi \quad (2.3)$$

$$\mathbf{e}_\theta \times \mathbf{e}_\phi = \mathbf{e}_r \quad (2.4)$$

$$\mathbf{e}_\phi \times \mathbf{e}_r = \mathbf{e}_\theta \quad (2.5)$$

$$D_r \mathbf{e}_r = D_r \mathbf{e}_\theta = D_r \mathbf{e}_\phi = 0 \quad (2.6)$$

$$\boxed{D_\theta \mathbf{e}_r = \mathbf{e}_\theta} \quad , \quad (2.7)$$

$$D_\theta \mathbf{e}_\theta = -\mathbf{e}_r \quad (2.8)$$

$$D_\theta \mathbf{e}_\phi = 0, \quad (2.9)$$

$$D_\phi \mathbf{e}_\theta = \mathbf{e}_\phi \cos\theta \quad (2.10)$$

$$D_\phi \mathbf{e}_r = \mathbf{e}_\phi \sin\theta \quad (2.11)$$

$$D_\phi \mathbf{e}_\phi = \mathbf{e}_z \times \mathbf{e}_\phi \quad (2.12)$$

$$\mathbf{e}_r = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta) \quad (2.13)$$

$$D_\theta \mathbf{e}_r = (\cos\theta \cos\phi, \cos\theta \sin\phi, -\sin\theta) \quad (2.14)$$

$$D_\theta \mathbf{e}_r = (\cos\theta \cos\phi, \cos\theta \sin\phi, -\sin\theta) \quad (2.15)$$

$$\mathbf{e}_\theta = \frac{\partial \bar{r} / \partial \theta}{\| \quad \|} = (\cos\theta \cos\phi, \cos\theta \sin\phi, -\sin\theta) \quad (2.16)$$

$$x(\bar{x}) = \chi(r, \theta, \phi) \quad (2.17)$$

Operador gradiente en coordenadas esféricas

$$D_{\bar{x}} X = \mathbf{e}_r D_r \chi + \frac{\mathbf{e}_\theta}{r} D_\theta \chi + \frac{\mathbf{e}_\phi}{r \sin\theta} D_\phi \chi \quad (2.18)$$

$$RX = \bar{x} X(\bar{x}) = r \mathbf{e}_r \chi \quad (2.19)$$

$$P = \frac{\hbar}{i} D = \frac{\hbar}{i} \left[\mathbf{e}_r D_r + \frac{\mathbf{e}_\theta}{r} D_\theta + \frac{\mathbf{e}_\phi}{r \sin\theta} D_\phi \right] \quad (2.20)$$

$$\begin{aligned} P\chi &= \frac{\hbar}{i} \left[\frac{1}{r} \frac{\partial}{\partial r} r \right] \chi \\ R\chi &= r e_r \chi \end{aligned}$$

Operadores en
coordenadas
esféricas.

(2.21)

- Si: $P^2 = P.P$, demostrar que:

$$P^2\chi = -\hbar^2 \left[\left(D_r^2 + \frac{2}{r} D_r \right) + \frac{1}{r^2} \left(D_\theta^2 + \text{ctg}\theta D_\theta + \frac{1}{\text{sen}^2\theta} D_\phi^2 \right) \right] \chi \quad (2.22)$$

$$P^2\chi = \left(e_r D_r + \frac{e_\theta}{r} D_\theta + \frac{e_\phi}{r \text{sen}\theta} D_\phi \right) \cdot \left(e_r D_r \chi + \frac{e_\theta}{r} D_\theta \chi + \frac{e_\phi}{r \text{sen}\theta} D_\phi \chi \right) \quad (2.23)$$

$$e_r D_r \cdot e_r D_r \chi = D_r^2 \chi \quad (2.24)$$

$$e_r D_r \cdot \frac{e_\theta}{r} D_\theta \chi = 0 \quad (2.25)$$

$$e_r D_r \cdot \frac{e_\phi}{r \text{sen}\theta} D_\phi \chi = 0 \quad (2.26)$$

$$\frac{e_\theta}{r} D_\theta \cdot e_r D_r \chi = \frac{D_r}{r} \chi \quad (2.27)$$

$$\frac{e_\theta}{r} D_\theta \cdot \frac{e_\theta}{r} D_\theta \chi = \frac{D_\theta^2 \chi}{r^2} \quad (2.28)$$

$$\frac{e_\theta}{r} D_\theta \cdot \frac{e_\phi}{r \text{sen}\theta} D_\phi \chi = 0 \quad (2.29)$$

$$\frac{e_\phi}{r \text{sen}\theta} D_\phi \cdot e_r D_r \chi = \frac{D_r}{r} \chi \quad (2.30)$$

$$\frac{e_\phi}{r \text{sen}\theta} D_\phi \cdot \frac{e_\theta}{r} D_\theta \chi = \text{ctg} \frac{D_\theta}{r^2} \chi \quad (2.31)$$

$$\frac{e_\phi}{r \text{sen}\theta} D_\phi \cdot \frac{e_\phi}{r \text{sen}\theta} D_\phi \chi = \frac{D_\phi^2 \chi}{r^2 \text{sen}^2\theta} \quad (2.32)$$

$$\boxed{P^2 \chi = -\hbar^2 \left[\left(D_r^2 + \frac{2}{r} D_r \right) + \frac{1}{r^2} \left(D_\theta^2 \operatorname{ctg} \theta D_\theta + \frac{1}{\sin^2 \theta} D_\phi^2 \right) \right] \chi} \quad (2.33)$$

Def: operador de momentum angular

$$\boxed{L = R \times P} \quad (2.34)$$

$$L \equiv (R_2 P_3 - R_3 P_2, R_3 P_1 - R_1 P_3, R_1 P_2 - R_2 P_1) \quad (2.35)$$

$$[R_j, L_k] = i \hbar \mathbf{e}_j \times \mathbf{e}_k \cdot \mathbf{R} \quad (2.36)$$

$$[P_j, L_k] = i \hbar \mathbf{e}_j \times \mathbf{e}_k \cdot \mathbf{P} \quad (2.37)$$

$$[L_j, L_k] = i \hbar \mathbf{e}_j \times \mathbf{e}_k \cdot \mathbf{L} \quad (2.38)$$

$$L_k = \varepsilon_{kij} R_i P_j \quad \begin{array}{c} \text{k} \curvearrowright \text{j} \\ \text{i} \end{array} \quad (2.39)$$

Definición

$$\boxed{L \times L = i \hbar L} \quad (2.40)$$

Momentum angular en coordenadas esféricas

$$L = \frac{\hbar}{i} (\mathbf{e}_\phi D_\theta - \csc \theta \mathbf{e}_\theta D_\phi) \quad (2.41)$$

$$\text{Demostrar que: } L^2 \chi = \left(\frac{\hbar}{i} \right)^2 (D_\theta^2 + \operatorname{ctg} \theta D_\theta + \csc^2 \theta D_\phi^2) \chi \quad (2.42)$$

$$\mathbf{e}_\phi D_\theta \cdot \mathbf{e}_\phi D_\theta \chi = D_\theta^2 \chi \quad (2.43)$$

$$-\mathbf{e}_\phi D_\theta \cdot \csc \theta \mathbf{e}_\theta D_\phi \chi = 0 \quad (2.44)$$

$$-\csc \theta \mathbf{e}_\theta D_\phi \cdot \mathbf{e}_\phi D_\theta = -\csc \theta \mathbf{e}_\theta \quad (2.45)$$

$$(\mathbf{e}_z \times \mathbf{e}_\phi) D_\theta x = \cot \theta D_\theta x \quad (2.46)$$

$$\csc \theta \mathbf{e}_\theta D_\phi \cdot \csc \theta \mathbf{e}_\theta D_\phi x = \csc^2 \theta D_\phi^2 x \quad (2.47)$$

$$(\mathbf{e}_z \times \mathbf{e}_\phi) \cdot \mathbf{e}_\theta = \mathbf{e}_z \cdot (\mathbf{e}_\theta \times \mathbf{e}_\phi) = \mathbf{e}_z \cdot (-\mathbf{e}_r) = -\cos \theta \quad (2.48)$$

Entonces

$$L^2 \chi = \left(\frac{\hbar}{i} \right)^2 (D_\theta^2 + \cot \theta D_\theta + \csc^2 \theta D_\phi^2) \chi \quad (2.49)$$

Por lo tanto

$$P^2 \chi = -\hbar^2 \left(D_r^2 + \frac{2}{r} D_r \right) \chi + \frac{L^2}{r^2} \chi \quad (2.50)$$

Definamos:

$$P_r \equiv \frac{1}{2} (\mathbf{e}_r \cdot \mathbf{P} + \mathbf{P} \cdot \mathbf{e}_r) \quad (2.51)$$

$$D_q: P_r = \frac{\hbar}{i} \left(D_r + \frac{1}{r} \right) \quad (2.52)$$

$$\mathbf{P} = \frac{\hbar}{i} \left(\mathbf{e}_r D_r + \frac{\mathbf{e}_\theta}{r} D_\theta + \frac{\mathbf{e}_\phi}{r \sin \theta} D_\phi \right) \quad (2.53)$$

$$\text{Demostrar } P_r^2 = -\hbar^2 \left(D_r^2 + \frac{2}{r} D_r \right) \quad (2.53)$$

$$\begin{aligned} P_r^2 \chi &= -\hbar^2 \left(D_r + \frac{1}{r} \right) \left(D_r + \frac{1}{r} \right) \chi \\ &= -\hbar^2 \left[D_r^2 \chi + \frac{D_r \chi}{r} + \frac{D_r \chi}{r} + \frac{\chi}{r^2} \right] \\ &= -\hbar^2 \left[D_r^2 \chi + 2 \frac{D_r \chi}{r} + \frac{\chi}{r^2} \right] \end{aligned}$$

Reemplazando (2.52) en (2.50):

$$\rightarrow P^2 \chi = \left[P_r^2 + \frac{L^2}{r^2} \right] \chi \quad (2.59)$$

Los temas para las exposiciones son los siguientes:

1. Efecto Zeeman normal (considerando el spin)
2. Interacción radiación materia (considere la radiación como onda)
3. Efecto Stark para el nivel n
4. Aplicación de la teoría de perturbaciones dependiente del tiempo (regla de oro de Fermi).
5. Cálculo de la energía del estado fundamental usando teoría de perturbaciones.
6. Microscopio de efecto túnel
7. Otra aplicación de cuántica que le interese (como láser y computación cuántica).

Escoja un tema que le interese, haga un resumen y expóngalo la última semana de clases.

Esto corresponderá a la práctica 8. Deben presentar un resumen de su exposición. Tomen como referencia el libro de cuántica para ingenieros de Fernández, el libro de Cohen, química cuántica de Levine u otros...

$$\boxed{H = \frac{1}{2\mu} \left[P_r^2 + \frac{L^2}{r^2} \right] + V} \quad (2.60)$$

$$V \chi(r, \theta, \phi) = V(r) \chi(r, \theta, \phi) \quad (2.61)$$

$$P_r = \frac{\hbar}{i} \left(D_r + \frac{1}{r} \right)$$

$$\boxed{P_r^2 = -\hbar^2 \left(D_r^2 + \frac{2}{r} D_r \right)}$$

$$V = -\frac{c}{r} \quad , \quad c > 0 \quad (2.62)$$

$$\chi(r, \theta, \phi) = n(r) k(\theta, \phi) \quad (2.63)$$

$$\frac{1}{2\mu} P_r^2 n k + \frac{L^2 n k}{2\mu r^2} + (V - E) n k = 0 \quad (2.64)$$

$$\frac{1}{n k} \left(\frac{k}{2\mu} P_r^2 n + \frac{n L^2 k}{2\mu r^2} + (V - E) n k \right) = 0 \quad (2.65)$$

$$\frac{P_r^2 n}{2\mu n} + \frac{L^2 k}{2\mu r^2 k} + \frac{(V - E) n}{n} = 0 \quad (2.66)$$

$$2\mu r^2 \left[\underbrace{\frac{P_r^2 n}{2\mu n} + \frac{(V - E) n}{n}}_{-\lambda \hbar^2} \right] + \underbrace{\frac{L^2 k}{k}}_{\lambda \hbar^2} = 0 \quad (2.67)$$

$$L^2 k = \lambda \hbar^2 k \Rightarrow \boxed{\lambda = \ell(\ell + 1)} \quad (2.68)$$

$$\ell \in \mathbb{Z}^+ \cup \{0\} \quad (2.69)$$

$$\frac{P_r^2 n}{2\mu} + (V - E) n = -\frac{\lambda \hbar^2 n}{2\mu r^2} \quad (2.70)$$

Ecuación radial

$$\frac{P_r^2 n}{2\mu} + \left(V + \frac{\lambda \hbar^2}{2\mu r^2} \right) n = E n$$

(2.71)

$$\frac{P_r^2 n}{2\mu} + \left(-\frac{c}{r} + \frac{\ell(\ell+1)\hbar^2}{2\mu r^2} \right) n = E n$$

$$\alpha^2 \equiv \frac{\hbar^2}{2\mu} :$$

$$-\alpha^2 \left(D_r^2 + \frac{2}{r} D_r \right) n - \frac{c}{r} n + \ell(\ell+1) \frac{\alpha^2}{r^2} n = E n$$

$$\text{Ensayemos } n = \frac{G}{r} :$$

$$D_r \frac{G}{r} = \frac{D_r G}{r} - \frac{G}{r^2}$$

$$\left\{ \begin{aligned} D_r^2 \frac{G}{r} &= \frac{D_r^2 G}{r} - \cancel{\frac{D_r G}{r^2}} + \cancel{\frac{2G}{r^3}} - \cancel{\frac{D_r G}{r^2}} \\ \frac{2}{r} D_r \frac{G}{r} &= \cancel{\frac{2}{r^2} D_r G} - \cancel{\frac{2}{r^3} G} \end{aligned} \right.$$

$$\Sigma = \frac{D_r^2 G}{r}$$

$$\left(D_r^2 + \frac{2}{r}D_r\right)G = \frac{D_r^2 G}{r}$$

$$\Rightarrow -\alpha^2 D^2 G + \left[-\frac{c}{r} + \ell(\ell+1)\frac{\alpha^2}{r^2}\right]G = EG$$

Veamos el comportamiento de G cerca al origen y en el ∞ :

$$\underline{r \approx 0}: -\alpha^2 D^2 G_0 + \ell(\ell+1)\frac{\alpha^2}{r^2}G_0 = 0$$

Ensayemos $G_0 = r^v$:

$$DG_0 = v r^{v-1}$$

$$D^2 G_0 = v(v-1)r^{v-2}$$

$$-\alpha^2 v(v-1)r^{v-2} + \ell(\ell+1)\alpha^2 r^{v-2} = 0$$

$$\ell(\ell+1) - v(v-1) = 0$$

$$\Rightarrow \begin{matrix} v = -\ell \\ v = \ell+1 \end{matrix} \quad \times \quad \leftarrow \begin{array}{l} \text{Para evitar la} \\ \text{singularidad} \\ \text{(Ejm: si } r = 0) \end{array}$$

$$\rightarrow \boxed{G_0 = r^{\ell+1}}$$

$$\underline{r \approx \infty}: -\alpha^2 D^2 G_\infty = EG_\infty$$

Ondas planas

Dispersión $G_\infty = e^{ikr}$

$$\rightarrow \alpha^2 k^2 e^{ikr} = E e^{ikr}$$

$$\rightarrow E = \alpha^2 k^2$$

Ensayemos: $\boxed{G_\infty = e^{-\sigma r}}$

$$-\alpha^2 \sigma^2 e^{-\sigma r} = E e^{-\sigma r}$$

$$E = -\alpha^2 \sigma^2 < 0$$

$$G_\infty = C_1 e^{-\sigma r} + C_2 e^{\sigma r}$$

ya que si
 $r \rightarrow \infty$
 se hace infinito

Ensayemos:

$$G(r) = G_0(r) F(r) \cdot G_\infty(r)$$

La ecuación para F es de la forma:

$$0 = F'' + 2 \left[\frac{(\ell+1)}{r} - \sigma \right] F' - \left[\frac{c}{r \alpha^2} + 2\sigma \frac{(\ell+1)}{r} \right] F$$

{ TAREA }

$$\sigma^2 = -\frac{E}{\alpha^2}$$

$$E < 0$$

$$F(r) = \sum_{k=0}^{\infty} a_k r^{k+S} :$$

$$0 = \sum_{k=0}^{\infty} a_k (k+S)(k+S-1) r^{k+S-2} + 2(\ell+1) \sum_{k=0}^{\infty} a_k (k+S) r^{k+S-2} - 2\sigma \sum_{k=0}^{\infty} a_k (k+S) r^{k+S-1} + \frac{c}{\alpha^2} \sum_{k=0}^{\infty} a_k r^{k+S-1} - 2\sigma(\ell+1) \sum_{k=0}^{\infty} a_k r^{k+S-1}$$

$$\begin{aligned}
0 &= a_0 S(S-1)r^{S-2} + 2(\ell+1)a_0 S r^{S-2} + \\
&\sum_{k=0}^{\infty} a_{k+1} (k+1+S)(k+S)r^{k+S-1} + \\
&2(\ell+1) \sum_{k=0}^{\infty} a_{k+1} (k+1+S)r^{k+S-1} \\
&- 2\sigma \sum_{k=0}^{\infty} [\quad] r^{k+S-1} + \frac{c}{\alpha^2} \sum_{k=0}^{\infty} a_k r^{k+S-1}
\end{aligned}$$

$(k+\ell+1)a_k$

Si S = 0:

Relación de recurrencia:

$$a_{k+1} = \frac{2\sigma(k+\ell+1) - \frac{c}{\alpha^2}}{(k+1)(k+2+2\ell)} a_k \quad k = 0, 1, 2, \dots$$

La serie anterior es convergente a la función: $e^{2\sigma r}$

($k \rightarrow \infty$)

$$G(r) \rightarrow r^{\ell+1} e^{2\sigma r} \quad \text{color: red } e^{-\sigma r}$$

No es una función acotada

Si k = n:

$$\sigma = \frac{c}{2\alpha^2(n+\ell+1)}$$

$$a_{k+1} = \frac{2\sigma(k+\ell+1) - 2\sigma(n+\ell+1)}{(k+1)(k+2+2\ell)} a_k$$

$$a_{k+1} = \frac{2\sigma(k-n)}{(k+1)(k+2+2\ell)} a_k$$

$$E = -\alpha^2 \sigma^2 = \frac{-\alpha^2 c^2}{4\alpha^4 (n+\ell+1)^2}$$

$$E = -\frac{c}{4\alpha^2} \frac{1}{N^2}$$

$$= -\frac{\mu c^2}{2\hbar^2} \frac{1}{N^2}$$

$$N = n + \ell + 1$$

$$E_n = -\frac{\mu c^2}{2\hbar^2} \frac{1}{N^2}; \quad N = n + \ell + 1$$

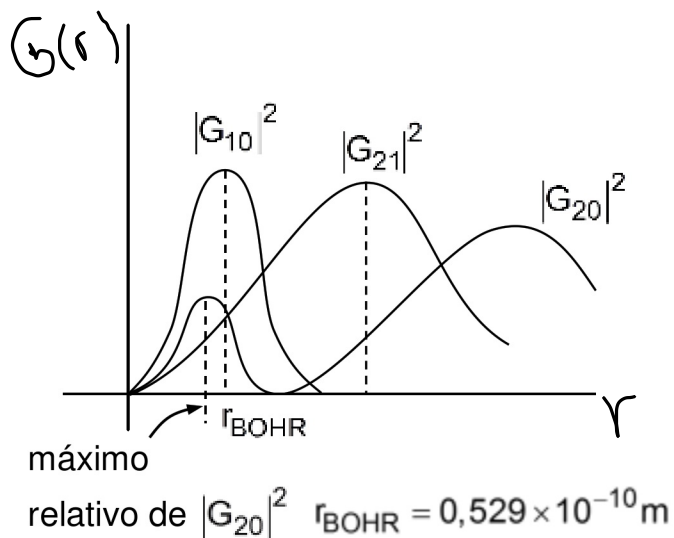
$$\begin{array}{ccccccc} \ell & 0 & 1 & 2 & \dots & N-1 \\ n & N-1 & N-2 & N-3 & \dots & 0 \end{array}$$

$$0 \leq \ell \leq N-1$$

$$\text{Si } \int |n(r)y_{(\theta,\varphi)}|^2 r^2 dr d\Omega$$

$$= \int_{-\infty}^{\infty} |G|^2 dr \underbrace{\int |y|^2 d\Omega}_1 = 1$$

$$\int_{-\infty}^{\infty} |G_{N\ell}|^2 dr = 1$$



Graficar las densidades de probabilidad radial $ G_{N\ell} ^2$:	N	ℓ	n	$G_{N\ell}$
	1	0	0	$G_{10} =$
	2	0	1	$G_{20} =$
		1	0	$G_{21} =$
	3	0	2	$G_{30} =$
		1	1	$G_{31} =$
		2	0	$G_{32} =$

Nota:

El número de máximo que tiene $|G_{N\ell}|^2$ es igual a $N - \ell$.

$$G_{N\ell}(r) = F_{N\ell}(r)r^{\ell+1}e^{\sigma r}$$

$$n_{N\ell}(r) = \frac{G_{N\ell}(r)}{r} = F_{N\ell}(r)r^{\ell}e^{-\sigma r}$$

$$\psi_{N\ell m} = n_{N\ell} V_{\ell m}$$

$$0 \leq \ell \leq N-1$$

$$-\ell \leq m \leq \ell$$

$$-\ell, -\ell+1, \dots, \ell-1, \ell$$

$$H, L^2, L_Z$$

Conjunto completo de observables que conmutan

$$[H, L^2] = 0$$

$$[H, L_Z] = 0$$

$$[L^2, L_Z] = 0$$

$$H\psi_{N\ell m} = -\frac{\mu c^2}{2\hbar^2} \frac{1}{N^2} \psi_{N\ell m}$$

$$L^2\psi_{N\ell m} = \ell(\ell+1)\hbar^2\psi_{N\ell m}$$

$$L_Z\psi_{N\ell m} = m\hbar\psi_{N\ell m}$$

$$N, \ell, m = m\hbar\psi_{N\ell m}$$

Los números cuánticos están asociados a los vectores propios de un conjunto de observables físicos medibles simultáneamente que describen a un sistema físico.

